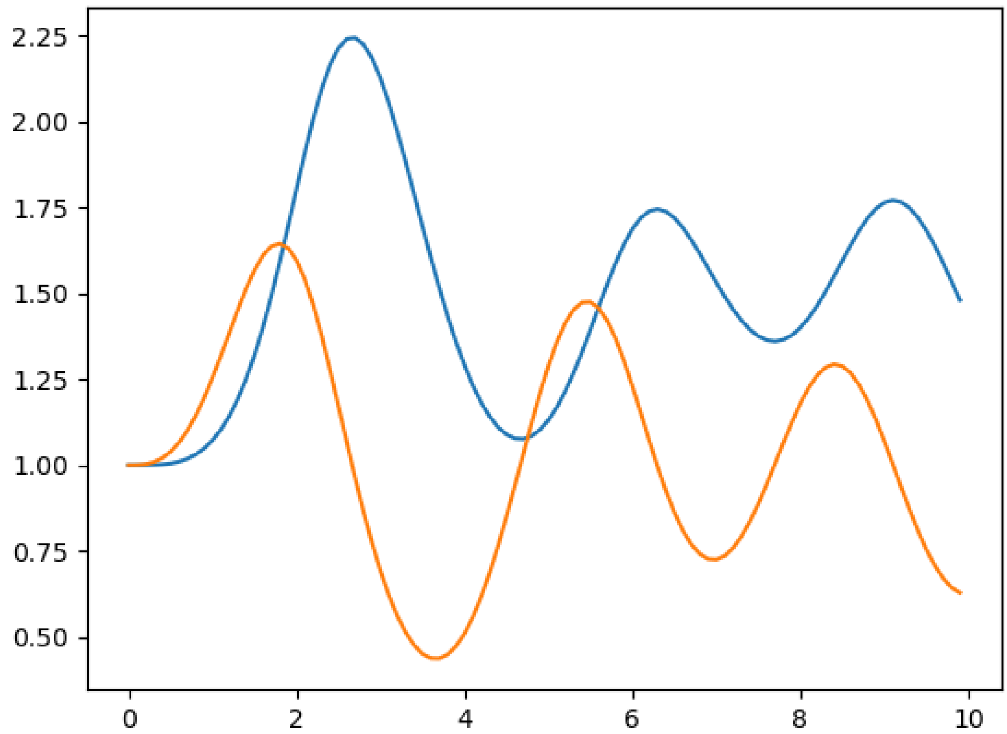
day 26 | 251029 W

In [1]:
```
%matplotlib ipympl

import numpy as np
import matplotlib.pyplot as plt
```

In [5]:
```
# define the differential equation
def f(r, t): # don't forget r is a vector!
    x = r[0]
    y = r[1]
    w = 1.0
    fx = x*y-x  # this is the dx/dt equation!
    fy = y - x*y + np.sin(w*t)**2  # this is the dy/dt equation!
    return(np.array([fx, fy]))

# define the boundary condition
a = 0.0
b = 10.0
N = 100
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([1.0, 1.0]) # initial condition

tpoints = np.arange(a, b, dt)

xpoints = []
ypoints = []

for i in tpoints:
    xpoints.append(r[0])
    ypoints.append(r[1])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6

fig0, ax0 = plt.subplots()
ax0.plot(tpoints, xpoints)
ax0.plot(tpoints, ypoints)
```

Out[5]:
```
[<matplotlib.lines.Line2D at 0x7667bf2964b0>]
```

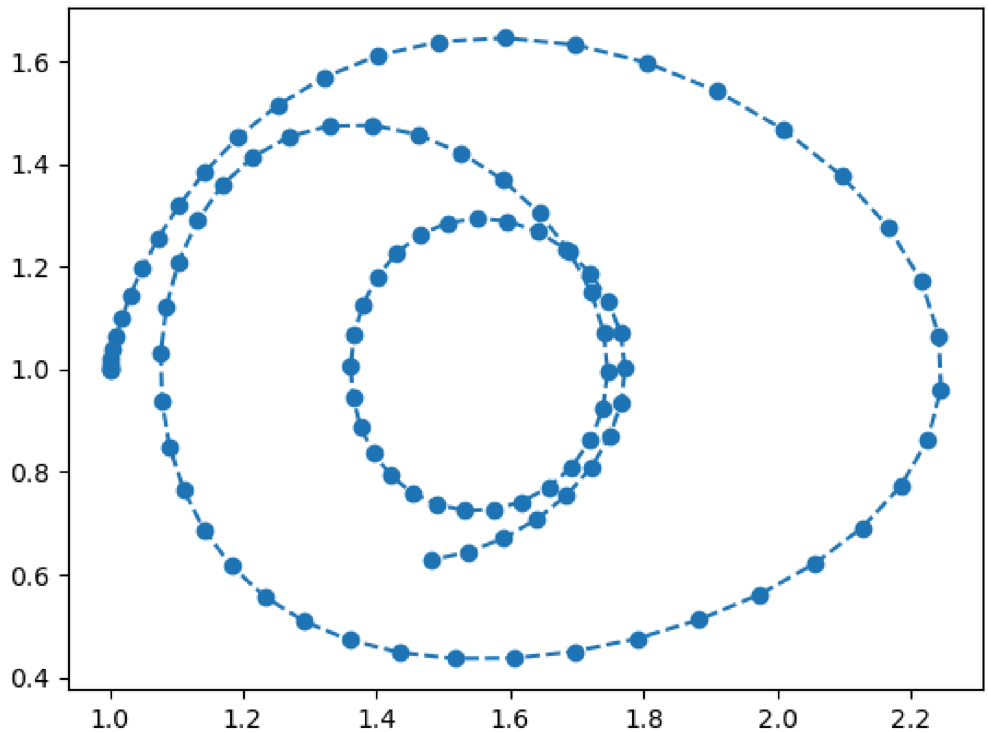
Figure



In [7]:
```
fig1, ax1 = plt.subplots()
ax1.plot(xpoints, ypoints, 'o--')
```

Out[7]:
```
[<matplotlib.lines.Line2D at 0x7667bc9e77a0>]
```

Figure



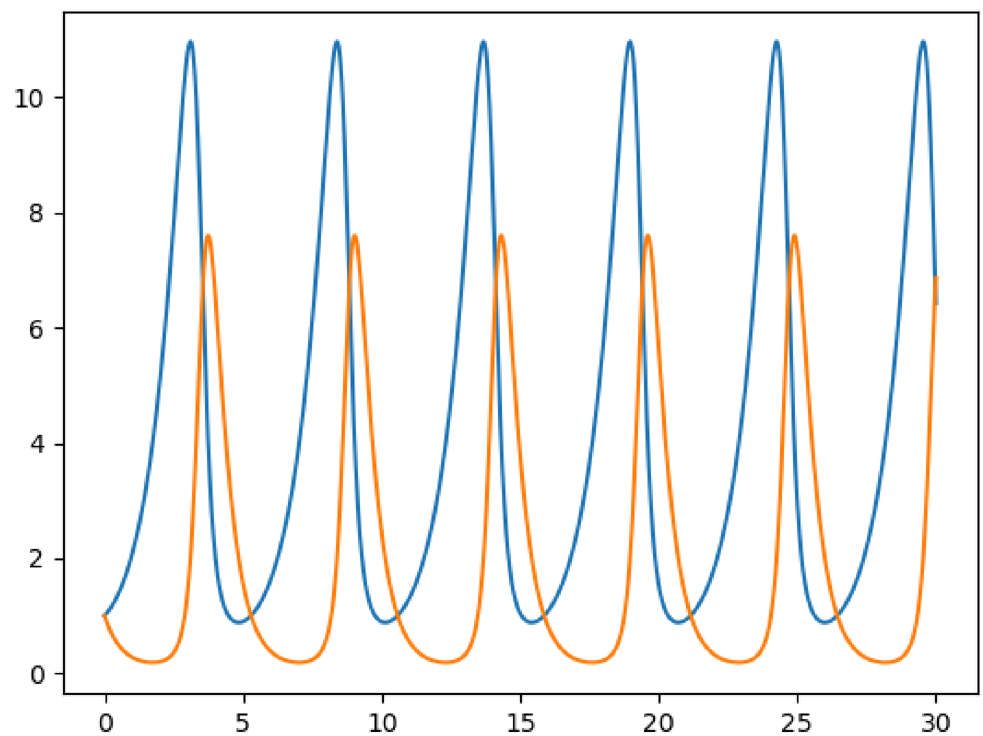
In [9]:

```
# define the differential equation
def f(r, t): # don't forget r is a vector!
    x = r[0]
    y = r[1]
    a = 1
    b = 0.5
    g = 0.5
    d = 2
    fx = a*x-b*x*y  # this is the dx/dt equation!
    fy = g*x*y-d*y  # this is the dy/dt equation!
    return(np.array([fx, fy]))

# define the boundary condition
a = 0.0
b = 30.0
N = 10000
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([1.0, 1.0]) # initial condition

tpoints = np.arange(a, b, dt)

xpoints = []
ypoints = []
```

```python
for i in tpoints:
    xpoints.append(r[0])
    ypoints.append(r[1])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6

fig2, ax2 = plt.subplots()
ax2.plot(tpoints, xpoints)
ax2.plot(tpoints, ypoints)
```

Out[9]: [<matplotlib.lines.Line2D at 0x7667bf3710d0>]

Figure



In [ ]:

In [14]:

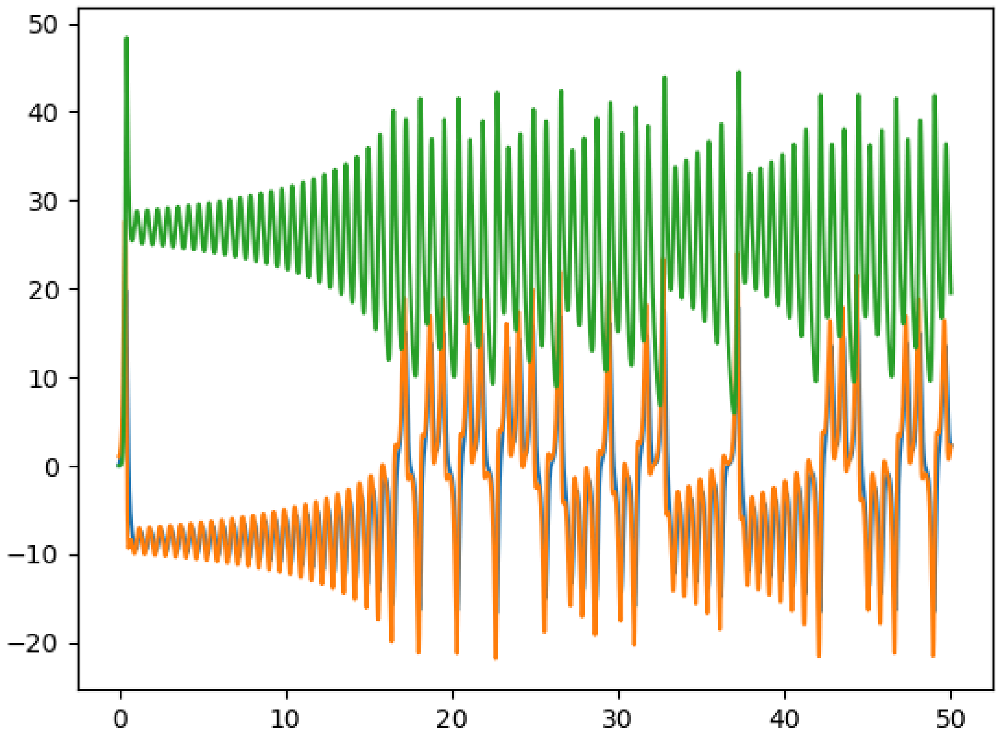
```python
# define the differential equation
def f(r, t): # don't forget r is a vector!
    x = r[0]
    y = r[1]
    z = r[2]
    sigma = 10
    rho = 28
    beta = 8/3.
    fx = sigma*(y-x) # this is the dx/dt equation!
    fy = rho*x - y - x*z  # this is the dy/dt equation!
```

```
    fz = x*y - beta*z # this is the dz/dt equation
    return(np.array([fx, fy, fz]))

# define the boundary condition
a = 0.0
b = 50.0
N = 10000
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([0.0, 1.0, 0.0]) # initial condition

tpoints = np.arange(a, b, dt)

xpoints = []
ypoints = []
zpoints = []

for i in tpoints:
    xpoints.append(r[0])
    ypoints.append(r[1])
    zpoints.append(r[2])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6

fig4, ax4 = plt.subplots()
ax4.plot(tpoints, xpoints)
ax4.plot(tpoints, ypoints)
ax4.plot(tpoints, zpoints)
```

Out[14]:

```
[<matplotlib.lines.Line2D at 0x7667bc5b25a0>]
```

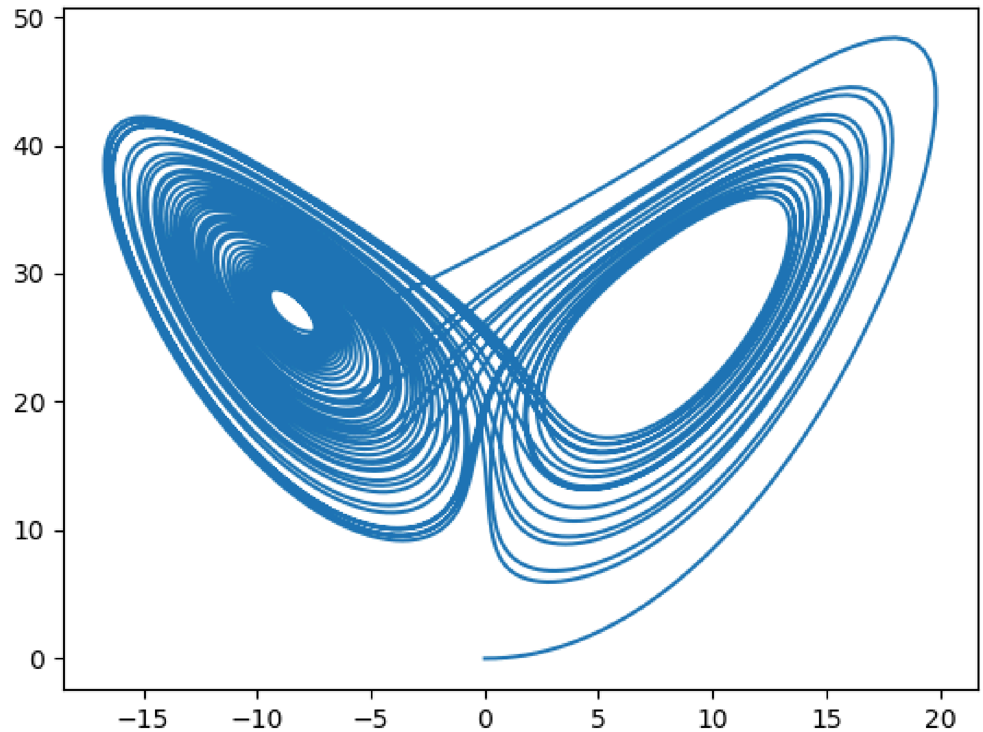
Figure



In [15]:
```
fig5, ax5 = plt.subplots(projection=)
ax5.plot(xpoints, zpoints)
```

Out[15]: [<matplotlib.lines.Line2D at 0x7667bc451c70>]

Figure




```
In [ ]:
```